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Toward Designing a Li-Fi-Based Hierarchical IoT Architecture

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ABSTRACT Current wireless communication technologies, such as wireless-fidelity (Wi-Fi) and Bluetooth, use radio waves as the basic medium in transferring information. Even though these technologies are widely spread, the need of exploring alternatives to transmit data wirelessly and more efficiently is a must. The reason behind that relies on the current limitations of the radio frequency (RF) band, which include overcrowding and interference with other RF applications. To explore alternatives, much research has been conducted to prove the possibility of using visible light as a wireless medium for transferring data. As a result, a new technology was presented by a German physicist Harald Haas, called light-fidelity (Li-Fi), which is a wireless technology that utilizes visible light as a communication medium instead of using the basic radio wave. Recently, Li-Fi technology has attracted the research community. Various studies have been conducted to improve the technology. However, there is still a noticeable need to support the research field due to the modernity of the technology. Hence, this paper presents an extensive survey of the previous studies and projects conducted on the technology, besides multiple leading companies working on the manufacture of Li-Fi-compatible products. In addition, a Li-Fi-based IoT architecture is proposed in this paper, which relies on the collection of data from multiple environments, where Li-Fi is installed. Li-Fi-generated data are analyzed and processed to make intelligent decisions to enhance services in many sectors.

INDEX TERMS Visible light communication (VLC), light-fidelity (Li-Fi), light emitting diodes (LEDs), photo-detector, Internet of Things.

I. INTRODUCTION

Wireless-fidelity (Wi-Fi), and many other wireless communication technologies used meanwhile, operate in the radio frequency (RF) band. However, due to the crisis of the overcrowded RF band, which ranges from interference, limited bandwidth to fear for human safety, exploring efficient alternatives is a must [1].

The RF band is just a small portion of the electromagnetic spectrum available for data transfer. A German physicist Harald Haas proved the possibility of using the visible light band for data transmission. He introduced the technology of light-fidelity (Li-Fi), which is based on the visible light communication (VLC) technology. It is a short-range wireless communication system that leverages the light generated by light emitting diode (LED) bulbs as the data transmission medium [2].

Li-Fi can contribute much more than the existing technologies using RF as a signal carrier (e.g., Wi-Fi, Bluetooth, and radio frequency identification). It is a preceding technology

with high data transmission rates, safety for human use, and security [1]. Moreover, the integration of illumination and communication services from the same source reduces the infrastructure complexity and energy consumption significantly [2].

Recently, the Li-Fi technology has attracted the research community, addressing the promising potential of the integration of Li-Fi into basic lightings. Many studies have been conducted to improve the technology and allow it to be used by people.

The Li-Fi industry is expected to grow drastically in the coming years. Many global lighting solution manufacturers are shifting their business to more service-oriented lighting solutions instead of basic off-the-shelf products. The Li-Fi products have experienced noticeable improvements in the market, and several products are being sold with increasing data rates and easy installations.

Li-Fi is considered as an appealing Internet of Things (IoT) application since it works in transferring basic lamps to access

points capable of receiving and transmitting data. Integrating Li-Fi in every light source creates an opportunity of shifting ubiquitous light to the connected objects.

Therefore, this paper provides a comprehensive survey of the technology. It covers the background information required to understand it, besides the research and applications conducted and developed to support its growth. Different companies around the world working on the research and development of this technology are presented in this survey. Furthermore, a comparison between multiple Li-Fi mobile applications has been conducted to reveal the limitations and issues that require improvements when developing Li-Fi-compatible mobile applications. The concepts of IoT and Li-Fi are integrated to propose a multilayered Li-Fi-based IoT architecture for indoor and outdoor deployment of Li-Fi systems. The architecture relies on the collection of data from multiple environments where Li-Fi is installed. Li-Fi-generated data are analyzed and processed to build an intelligent decision-making system to enhance services in many sectors.

This paper has the following contributions:

- A comprehensive survey on the Li-Fi technology, including background information required to understand it, in addition to a walkthrough of the research conducted on the technology to support its development.
- Highlighting various industrial efforts from companies around the world to embrace Li-Fi-compatible solutions and products.
- The proposal of a Li-Fi-based IoT hierarchical architecture to develop an IoT system that collects Li-Fi data generated from various locations. The proposed architecture aims to analyze the collected data and build smart decisions upon it.

The remainder of the paper is organized as follows: section II provides background information on the Li-Fi technology, its features, the basic architecture, its working principle, and its limitation. Section III presents a walkthrough of the research conducted on the Li-Fi technology and how it evolved through the years; a list of significant research has been provided at the end of the section. Section IV discusses the Li-Fi industrial trends over the recent years and the availability of Li-Fi-compatible commercial products. Various products, projects, and applications are presented in this section. In addition, the section presents a comparison between the currently developed applications supporting Li-Fi to reveal the limitations that require improvements and the features needed to achieve the proposed architecture. Section V proposes a hierarchical Li-Fi-based IoT architecture in the aid of developing a network of interconnected lights. The last section concludes the paper and presents future work.

II. BACKGROUND

A. LIGHT FIDELITY TECHNOLOGY

Li-Fi technology is a VLC technology that employs LED bulbs for illumination and data transmission.

VLC is a version of data communication systems that leverages visible light for communication. It occupies the band from 380 nm to 750 nm in the electromagnetic spectrum, which corresponds to a frequency spectrum of 430–790 THz. Since the visible light spectrum is 10,000 times larger than the RF spectrum, VLC is regarded as a solution to RF bandwidth limitations [3].

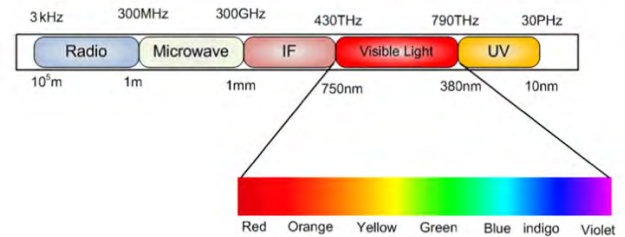


FIGURE 1. VLC frequency spectrum [3].

The Li-Fi technology overcomes several inefficiencies of the current wireless communication technologies using the RF band (e.g., Wi-Fi, IrDA, and Bluetooth). One of the biggest struggles of the RF band is “spectrum crunch” [4], which is the incapability to keep up with the huge demand of wireless communication and high data rates due to the limited bandwidth [4]. Moreover, interference sufferings, low data rates, and human safety concerns are other significant downgrades of the RF band. Thus, exploring alternatives is a must, and Li-Fi proves to be a suitable candidate for this purpose. It can be seen as a preceding technology when compared with other wireless technologies from the following viewpoints:

1) CAPACITY

Since Li-Fi works with the visible light spectrum, it provides 10,000 times more license-free bandwidth of the entire RF spectrum [5].

2) SPEED

Li-Fi can transmit data at a peak speed of more than 1 Gbps, which is estimated to reach 10 Gbps, due to low interference, exceeding all current RF technologies (Wi-Fi of 150 Mbps, Bluetooth of 3 Mbps, and IrDA of 4 Mbps) [6]. It should be mentioned that the traditional Shannon framework is not applicable for computing the transmission data rate.

3) EFFICIENCY

The core technology of Li-Fi is LED, and hence, it is more efficient in terms of energy consumption and construction cost [4].

4) AVAILABILITY AND SAFETY

Li-Fi is an eco-friendly technology and does not possess any health concerns since it uses visible light as the communication medium, unlike RF technologies, which have safety as a major concern. For instance, hospitals prohibit using Wi-Fi in some areas such as operation theaters, as it interferes

with other devices, blocking the monitoring equipment's signals [6].

5) SECURITY

"If you can't see the light, you can't access the data!" [4]. Light waves are incapable of penetrating walls, which makes them invulnerable to interception and misuse.

B. ARCHITECTURE OF Li-Fi

The architecture of the Li-Fi system relies on two essential optoelectronic components: LED bulbs and photo-detectors (PDs) [6]. PDs and other components necessary for data reception are integrated into dongles, called a Li-Fi dongle.

1) LIGHT EMITTING DIODE (LED)

The heart of the Li-Fi technology is a high-brightness LED bulb. LED is a semiconductor device that emits light when an electric current is passed through it [7].

In case of Li-Fi, LEDs act as data transmitters. Any basic LED bulb can be used for this purpose, the only important factor being that it must be equipped with a microchip so that it can illuminate and transmit data wirelessly at the same time [6]. The ubiquitousness of lighting and the shifting toward using LED bulbs for illumination, instead of standard incandescent and compact fluorescent lamps, are additional advantages of Li-Fi, creating an existing infrastructure that reduces cost and complexity.

2) Li-Fi DONGLE

A Li-Fi dongle receives the transmitted data. It comprises a PD, an amplifier, and a data processing circuitry.

PD: or photosensors are sensors of light or other electromagnetic energy that convert light photons into current. In case of Li-Fi, PDs are equipped in "dongles," and hence, they can be attached to digital devices such as PCs, laptops, and mobile phones, allowing them to act as data receivers.

Amplifiers and data processors: they are used to increase the power of the signal received from the PD and process it to eliminate any noise that might affect the signal.

C. Li-Fi WORKING PRINCIPLE

The significant concept behind the Li-Fi technology is wireless data transmission through illumination [8]. The working principle of Li-Fi is quite similar to that of Wi-Fi, as both transmit and receive data electromagnetically. In general, to obtain a working Li-Fi system, you need to have a data transmitter on one end (i.e., LED light) and a data receiver on the other end (i.e., PD). LEDs selection plays essential role as the data rate in a Li-Fi system can be related with LEDs. Parameters like ON-OFF speed and number of LEDs can affect the data of the communication. If ON-OFF speed of LED is fast, then data can be transmitted at high rates in the form of 1's and 0's. Higher the number of LED's in a system results in more transmission of data.

1) DATA TRANSMITTER – LED

Encoding data into light for transmission is achieved by varying the light intensity of the LED, which causes the LED to flicker ON and OFF at a very high speed [5]. These flickers represent the data being transmitted. When the LED is switched ON, it logically represents the transmission of a "1," and when it is switched OFF, it transmits a "0." A combination of 1s and 0s generates different data strings. However, the flickering speed is so high that the LED appears to be constant to the human eye, causing no harm to it [9].

2) DATA RECEIVER – DONGLE

Data reception is the function of the dongle, or more specifically, the PD within. A PD detects the light source and converts the changes in its intensity into an electrical signal. When the LED flickers ON, the PD registers a binary 1; otherwise, it registers a binary 0. Then, the received data are amplified, processed, and forwarded to the user [10].

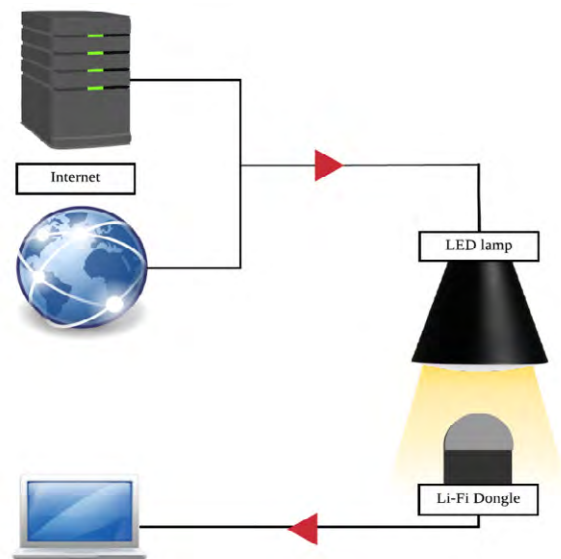


FIGURE 2. Block diagram of Li-Fi system [10]

D. Li-Fi LIMITATIONS

Li-Fi is a promising technology, but like any other emerging technology, it has its downsides. Following are the major limitations of Li-Fi [11]

- 1) It requires a perfect line-of-sight to receive data.
- 2) Light cannot pass through objects; thus, if the receiver is blocked by any opaque object, then the signal is immediately cut out.
- 3) There is interference from other light sources such as sunlight and normal bulbs. However, most new Li-Fi products have overcome this limitation, since Li-Fi depends on detecting the fast changes in light intensity and not on the slowly varying levels caused by natural disruptions in daylight or sunlight. A team from PureLiFi company has tested

the company's Li-Fi receivers outdoor under 77,000 Lux of sunlight.

4) It is a short-range wireless communication system. However, professor Harald Hass has mentioned that huge improvements in data rates, ranges and coverage have been reported [15].

5) To be able to transmit data, light need to remain ON all the time. However, according to Harald Hass, the light can provide high data rate even if it is operated at very low level.

The above limitations can be considered as issues or challenges that Li-Fi technology can introduce compared with other technology such as cellular communication and Wi-Fi.

III. Li-Fi RESEARCH

Communicating through visible light has a long history, even though the current LED-based VLC technologies were invented a few years ago. The early years of VLC were merely based on the use of sunlight, fire, or any type of lamp for transmitting information [3]. The first data transmission through optical wireless communication dates back to 1880 in Washington, D.C., when Alexander Graham Bell invented the photophone. In the photophone experiment, he modulated sunlight to act as a means of carrying voice signals and transmitting it over a distance of around 200 m. However, the photophone experiment did not work very well due to severe interference and the use of an ordinary receiver [12].

It took a long time to develop components and technologies providing high data rates in VLC. The development of the LED enhanced the light intensities and modulation bandwidths for the VLC [12]. One of the preceding studies on using LEDs in the visible light spectrum to transmit data was conducted in 2001, when Twibright Lab's open-source project achieved a data rate of 10 Mbps using a red LED. Later in 2003, the visible light communication consortium was established in Japan, which was a significant step toward the evolution of LED-based VLC [12].

Since then, many research activities have been conducted by universities, companies, and organizations around the world that have focused on VLC. From the University of Edinburgh in the UK, professor Harald Haas has been working on the idea that data can be transmitted through LED bulbs. In 2006, professor Haas and his research group published the first proof-of-concept results demonstrating that commercially available LED bulbs can be turned into wireless transmission systems. They exploited the high peak-to-average ratio in orthogonal frequency division multiplexing (OFDM), which is considered as a disadvantage in RF transmission systems [13]. The paper's results proved that the OFDM techniques can be used to modulate the intensity of LEDs.

In 2010, professor Haas and his research group started working on the D-Light Project (data through illumination). According to a team member, the project's objective was to achieve a minimum of 100 Mbps data rate from a standard off-the-shelf LED bulb under normal lighting conditions [14].

The D-Light system used OFDM techniques, which made it possible to vary the intensity of the LEDs at a very fast rate, which cannot be detected by the human eye [4]. The D-Light project's objective was achieved successfully with 102.5 Mbps data measured under normal lighting conditions [4].

After years of research in the field, professor Hass introduced the term Li-Fi to the world in a TED global talk in 2011. He described how he became motivated to find an alternative to the current system of wireless data transmission, and showed how his Li-Fi wireless data transmission system through LED lighting solves issues in the current RF system. He basically used a table lamp with an LED bulb to transmit a video of blooming flowers, which was then projected onto a screen behind him. During the event, he periodically blocked the light from the lamp to prove that the lamp was indeed the source of incoming data. He demonstrated a data transmission rate of around 10 Mbps [15].

In a breakthrough, a research group from the universities of Edinburgh, St. Andrews, Strathclyde, Oxford, and Cambridge cooperated in a research project called Ultra-parallel Visible Light Communication (UP-VLC), which ran from October 2012 to February 2017. They achieved record data transmission speeds of 10 Gbps. The researchers used a micro-LED bulb to transmit separate 3.5 Gbps data via each of the three primary colors red, green, and blue, adding up to a combined 10 Gbps transfer rate. Using OFDM modulation techniques, the micro-LED acted as a fast ON/OFF switch by manipulating its light intensity, allowing it to transmit chunks of data in binary form (series of zeroes and ones) at a high speed [16].

In 2014, Fujitsu Laboratories developed a technology that can modulate the color of light emitted by the LED bulbs in a way that it is not detectable to the human eye but can embed ID data in the light that is cast on the object by using color modulation, as shown in Fig. 10, which is a key feature of this technology. As the LED colors combine the three lights of red, green, and blue to produce a range of colors, modulating the intensity of the light emitted by each of the three colors along the time axis makes it possible to embed ID data in the light cast onto the object. Data for one ID are attached to each LED light. The data are simply captured by pointing at the object with a smartphone, which then captures the data corresponding to the ID [17]. This technology can be used in stores to provide product information or in museums to guide the visitors.

In 2016, a research group developed an automatic billing system using the Li-Fi module, as shown in Fig. 11. The system aims to help saving the customers' time instead of standing in a long queue for the payment process. The system comprises four sections: product section, trolley section, customer mobile section, and gate section. In the product section, each product is attached to a Li-Fi module that contains the product ID and other product-related information. In the trolley section, the trolley is embedded with a Li-Fi transceiver for the transmission of the product information.

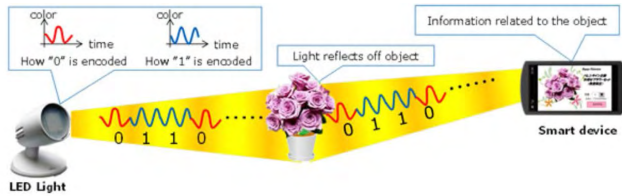


FIGURE 3. Color modulation used to transmit and receive data [30].

In the customer mobile section, the mobile is connected to a LiFi transceiver via a USB On-The-Go to retrieve the product ID and its price.

The system works as follows: the customer's mobile phone scans a product and displays the product ID, name, and price. When the product is placed in the trolley, which also has a Li-Fi module, its information is registered in the customer's shopping cart. After that the payment process is completed in the mobile app via the banking system, the bill details are sent to the gate module, which has a Li-Fi transceiver. At the gate, the trolley is scanned for final verification; if the payment process is fully complete, the gate will open, else an alarm gets triggered [18].

Table 1 presents the significant research conducted on Li-Fi.

IV. LI-FI INDUSTRY TRENDS AND STATUS

The emerging Li-Fi field has gained noticeable attention from not only researchers but also entrepreneurs, corporates, and startups that are willing to be part of what may be a giant market. The shift toward using eco-friendly LEDs is another genuine game-changer in the future of the Li-Fi market. In addition, its existing infrastructure makes investing in the technology more tempting. Deploying Li-Fi technology in every lamp provides illumination, location-based content, and an on-demand connectivity to the Internet. These features led an active group of new start-ups and large firms around the world to devote their efforts in the development of the Li-Fi technology, with some lighting solution providers even shifting their interest from traditional lighting toward providing smart lighting solutions. Li-Fi was an ideal investment for many global companies, enabling products, projects, and technologies. This section provides a list of global leading companies and their effort in the growth of the Li-Fi market.

A. Li-Fi PRODUCTS

Many global firms have been working toward the manufacture of Li-Fi ready-to-use commercial products. The Li-Fi products mainly fall into two categories depending on how data travel in the network: unidirectional and bidirectional products. In unidirectional products, data travel one way from the source to destination (i.e., LED to receiving device). Usually, this range of products is referred to as positioning systems providing location-based services.



FIGURE 4. Automatic billing system using Li-Fi module [18].

Unidirectional products are assisted with other means of wireless communications such as Wi-Fi, Bluetooth, or 3G/4G.

Bidirectional products, on the other hand, provide two-way communication: an uplink and a downlink. In this product range, the goal is to provide a fully functional wireless network system reliant on Li-Fi, allowing mobility with handover capabilities, multiple access, and high transmission speeds. The LED in bidirectional products acts as the network access point and must be connected to a backbone network using the Ethernet to provide networking capabilities. The dongle in this range is an essential component as it acts as the transmitter that sends the uplink. In the next subsections, three top companies in the Li-Fi market are presented with the Li-Fi-compatible products they released.

1) PureLiFi COMPANY

The first commercialized Li-Fi product available in the market was manufactured by PureLiFi, a company specialized in VLC technology called the "Li-1st" [19]. The product provides full duplex communication with a capacity of 11.5 Mbps (approximately 5 Mbps uplink and 5 Mbps downlink) and a coverage of 3 m. The product consists of two units: ceiling unit (i.e., the LED) and desktop unit (i.e., the dongle). The product is considered a legacy product and longer being sold.

An year later, PureLiFi released the Li-Flame, the first Li-Fi product to allow handover between LED access points. It also faced a noticeable improvement in the transmission data rates with a link capacity of 10 Mbps uplink and 10 Mbps downlink. This product is considered another legacy product. Following Li-Flame, one of the PureLiFi legacy products was the LiFi-X, which allows full-duplex, bidirectional communication but with an uplink and downlink speed of 42 Mbps. It also provides full mobility and multiple users.

At the beginning of 2018, PureLiFi showed more commitment in possessing the Li-Fi market by making significant improvements in its Li-Fi components and releasing its new generation of Li-Fi products, the LiFi-XC. The LiFi-XC system consists of an access point and a USB dongle. The solution supports mobility, multiple users' capabilities supporting up to 8 stations at a time and a full-duplex, and bidirectional communication with 43 Mbps link rate (43 Mbps uplink and 43 Mbps downlink). It also provides geolocation and geofencing services. The dongle supports Windows 10, Window 7, Linux, and Mac OS operating systems and has high

TABLE 1. Summary of previous studies.

Year	Author	Main contributions
2006 [13]	Prof. Harald Haas, Dr. Mostafa Afgani, Hany Elgala, and Dietmar Knipp	Working on the idea that data can be transmitted through LED bulbs. Demonstrating that it is possible to turn commercially available (LED) light bulbs into wireless transmission systems.
2010 [4]	Prof. Harald Haas, Dr. Mostafa Afgani and Dr. Wasiu Popoola	The D-Light project's objective was achieved successfully with 102.5 Mbps data transmission measured under normal lighting conditions.
2011 [15]	Prof. Harald Hass	Introduced the term Li-Fi to the world in a TED global talk. Companies and industry groups formed the Li-Fi alliance.
2012 [16]	Group of researchers from the universities of Edinburgh, St. Andrews, Strathclyde, Oxford and Cambridge.	Research project called UP-VLC. Achieved record data transmission speeds of 10 Gbps.
2014 [17]	Fujitsu Laboratories	Developed a technology that can modulate the color of the light emitted by the LEDs in a way that it can embed ID data in the light that is cast on the object.
2016 [18]	Zubin Thomas, Nikil Kumar and D. Jyothi Preshiya	Developed an automatic billing system using Li-Fi module.

performance optics with a 60° field of view. The LiFi-XC access point supports various LED fixtures and supports both power over Ethernet (POE) and power-line communication (PLC) options [20].

2) OLEDCOMM COMPANY

Another pioneer company in the making of Li-Fi products, is Oledcomm, a French company specialized in designing Li-Fi solutions for LED-based lighting systems. Oledcomm has been in the market since 2012 and has partnered with many global companies in lighting solutions, such as Philips, Thorn, and OSRAM. Oledcomm has been delivering its solutions distinctively from any other company in the industry. The company did not limit its solutions to ready-to-use Li-Fi products, but instead has been designing and delivering firmware, modems, chipset, and API/SDKs, targeting a wider range of consumers by enabling lighting engineers, system integrators, electrical installers, and software developers to benefit from the technology. Oledcomm has been providing its solution in partnership with many companies, by equipping its two-range Li-Fi modules in the products of other companies. Oledcomm products are all either equipped with Oledcomm's GEOLiFi® or LiFiNet® technology. The GeoLiFi product range is a unidirectional Li-Fi-enabling technology that adds accurate outdoor and indoor geolocation capabilities. LiFiNet, on the other hand, is a bidirectional point-to-point communication system in compliance with the IEEE 802.15.7r1 standard. Besides its various features and services, Oledcomm has been designing its solutions to be suitable for both indoor and outdoor environments.

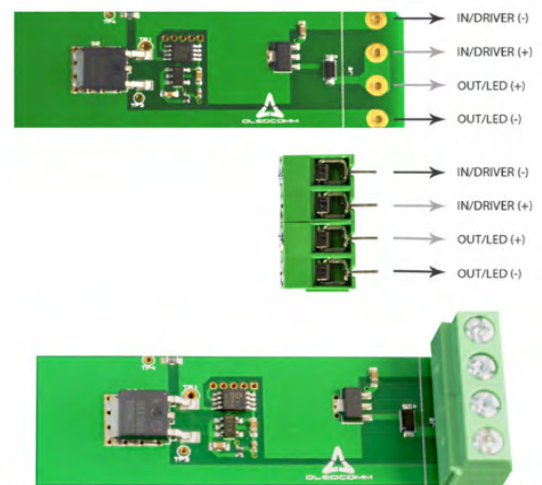
**FIGURE 5.** GeoLiFi transceiver module by Oledcomm [21].

Fig. 5 presents Oledcomm's GeoLiFi transceiver module, which is compatible with a wide range of LEDs and suits both indoor and outdoor illumination. The transceiver can be either integrated in the driver or connected externally to the driver and LED lamps. In both schemes, the outputs are connected to the LED driver, while the inputs are connected to the LED lights [21].

The LiFiNet technology can be installed using the LiFiNet kit by Oledcomm, which consists of multiple components creating the whole system. The kit is designed for lighting manufacturers willing to integrate bidirectional Li-Fi in their

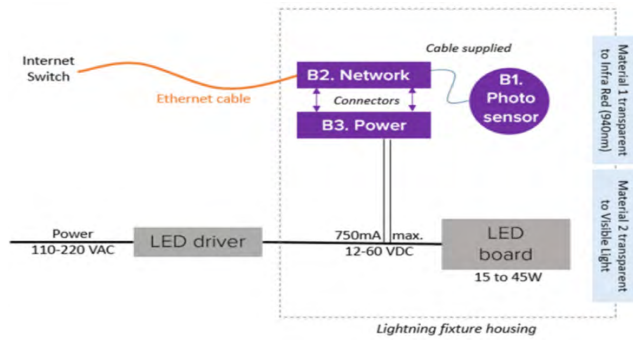


FIGURE 6. Architecture of LiFiNet by Oledcomm [22].

products. The integration creates an opportunity of transforming lighting solutions into an Internet source [22].

Fig. 6 illustrates a LiFiNet empowered system, where B1, B2, and B3 indicate the components light sensor board, power board, and network board, respectively, in the kit.

Oledcomm has partnered with Technilum, a company that designs and manufactures urban lighting furniture, to equip one of its streetlight product range with the GeoLiFi technology. The streetlight Li-Fi detection range covers a distance of up to 4 m and has a light beam angle of 120° [23]. In 2016, the district of Camille Claudel in Palaiseau, France, was installed with 77 of these streetlights. The project aimed to test the Li-Fi technology in urban context [24].

In 2018, Oledcomm has unveiled one of its latest innovations at CES 2018 (Consumer Electric Show) in Las Vegas, called MyLiFi, which is a desk lamp working as a bidirectional communication link delivering a high Internet speed of up to 23 Mbps (13 Mbps downlink and 10 Mbps uplink). However, a laboratory test has shown a theoretical speed of 224 Mbps. A customized mobile-based application has been designed to fully control the LED [25].

Oledcomm has a wide range of products alongside the mentioned ones. Most importantly, it provides a GeoLiFi discovery kit for those interested to test and discover the technology. GeoLiFi LED Tube, LiFiCare Lamp for clinics and hospitals, LiFiNet LED panel for offices, LudicLiFi floor lamp for offices, GeoLiFi tracker for trolleys, and LiFiNet streetlight are other significant Li-Fi products manufactured by Oledcomm's partners and equipped with Oledcomm's innovative technology [26].

3) VLNComm COMPANY

VLNComm is an American company working on the development of LED-based solutions with embedded Li-Fi capabilities. VLNComm participated in shaping IEEE 802.15.7 (VLC) standards. The company is drawn by the U.S Department of Energy and the National Science Foundation [27]–[29], and mainly focuses on delivering high-speed bidirectional communication through light. It provides two Li-Fi-enabling solutions and a dongle that is compatible with both. Compared to other companies, VLNComm has reached high data transmission rates with its product, i.e., up

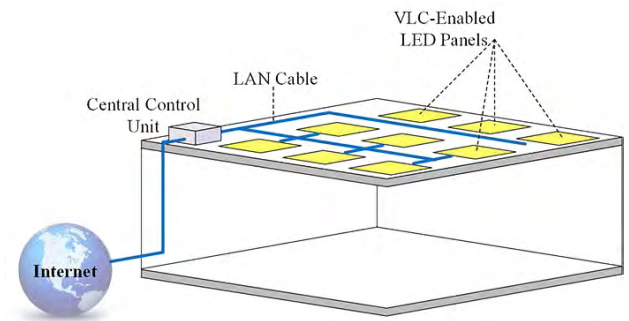


FIGURE 7. Architecture of LumiNex panel developed by VLNComm [31].

to 108 Mbps downlink speeds. The first solution VLNComm provides is a Li-Fi-enabling desk lamp called “LumiNet Desktop” [30], which can be connected to the Ethernet through its Ethernet port, delivering a speed of 23 Mbps uplink and 23 Mbps downlink. The lamp can connect up to seven users simultaneously. It has a coverage area of 1 m² and uses an RGB LED for data transmission. The other product is a ceiling solution called “LumiNex Panel,” which provides 108 Mbps download speed when coupled with the LumiStick 2 dongle. It has 53 Mbps upload speed and can cover an area of 516 square feet [29]. The solution can provide Internet up to 15 users with low latency (less than 0.4 ms). Fig. 7 illustrates the LumiNex solution in an indoor environment, where each panel is connected to the LAN through an Ethernet cable.

LumiStick 2 is a dongle designed by VLNComm to be compatible with both its products. The dongle uses a USB interface, has a field of view of 120°, and supports Windows, Mac OS, and Linux operating systems.

Fig. 8 shows the different Li-Fi products manufactured by Oledcomm, PureLiFi, and VLNComm.

As shown in Table 2, each company uses a specific modulation technique with its products depending on its specification and requirements. There are different data modulation techniques used for optical wireless communication, with Hadamard coded modulation (HCM) and OFDM being two significant ones. Unlike PureLiFi, which uses OFDM modulation techniques in its products, VLNComm uses HCM for data transmission in the optical wave. Laboratory experiments have proven that OFDM variations are more efficient in terms of power consumption levels in comparison to HCM, which was proposed as a solution to the limitation of OFDM modulation at higher illumination levels [32], [33].

Besides being an appealing technology that would significantly change the future of communication, Li-Fi is indeed a promising technology that brings many opportunities for the business sector as well. As presented, several companies are dedicating their work in the field of Li-Fi by developing and producing commercial Li-Fi products. Since an LED bulb forms the heart of the Li-Fi technology, it is reasonable to say that the technology has also brought a chance for the manufacturers of light bulbs to get involved in a new

TABLE 2. Li-Fi product specifications.

Product	Company	Data Rate	Modulation Technique	Receiver Spec.	Coverage Range	Communication Link Spec.	Other Notes
LiFi-XC [20]	PureLiFi	• 43 uplink • 43 downlink	OFDM	Li-Fi dongle as downlink receiver with USB connector empowered with infrared for uplink communication	2.8 m ²	• Full Duplex • Bidirectional	• Inter-AP handover • Multiple users support up to 8 stations
LiFi-X [19]	PureLiFi	• Downlink 42 Mbps • Uplink 42 Mbps	OFDM		-	• Full Duplex • Bidirectional	• Inter-AP handover • Multiple users
MyLiFi[25]	Oledcomm	• 13 Mbps downlink • 10 Mbps uplink	-	USB Li-Fi dongle	-	• Full Duplex • Bidirectional	-
LUDICLiFi Lamp [26]	Oledcomm	• 1 Mbps downlink • 1 Mbps uplink	-	USB Li-Fi dongle	1.8 m in line of sight	• Bidirectional	• One user connected at a time
GEOLiFi Street Lighting [23]	Oledcomm and Technilum	• 1 Mbps downlink • 1 Mbps uplink	-	Headphone Jack Li-Fi dongle or camera modulation. Depends on the equipped transceiver chip	Up to 4 m high LiFi detection	• Unidirectional	• One user connected at a time
LumiNet Pad [29]	VLNComm	• Downlink 108 Mbps • Uplink 53 Mbps	HCM	USB Li-Fi dongle empowered with Infrared for uplink communication. Wavelength band:	48 m ²	• Full Duplex • Bidirectional	• Multiple users support up to 15 stations
LumiNet Desktop [30]	VLNComm	• Downlink 23 Mbps • Uplink 23 Mbps	HCM		1 m ²	• Full Duplex • Bidirectional	• Multiple users support up to 7 stations

promising field. The adoption of Li-Fi as a service is an excellent opportunity to reinvent their businesses, especially those who are struggling in the market. In the future, the lighting business may become a service rather than just a basic product.

B. Li-Fi CASE STUDIES

1) CARREFOUR LILLE

In 2015, the Philips lighting solutions company cooperated with Carrefour Lille retail stores in France to install an

LED-based communication system. The system allows customers to find products throughout the store using a specially designed smartphone application. The system transforms shopping into an interactive experience and provides new services to Carrefour customers. The mobile application developed for Carrefour Lille intended to test a new service that goes beyond just lighting. The main focus of the application was to allow customers to create shopping lists and receive different promotions. Once the customer is in the store, he/she can easily locate all products in the shopping list by using the



FIGURE 8. Collection of Li-Fi commercialized products.

mobile application. The application is available for both iOS and Android. It is developed by Philips in cooperation with IBA and Carrefour store [34].

2) E. LECLERC RETAIL STORES

In 2016, Oledcomm cooperated with E. Leclerc retail store in France to develop an indoor positioning system and a Li-Fi-compatible mobile application. The store's LED lights are equipped with Li-Fi technology and the trolleys' handles have the Oledcomm's GeoLiFi tracker module, which registers the trolley path, as shown in Fig. 10. The geo-localization of the LEDs enables the system to accurately follow the customer's journey, which helps E. Leclerc owners to learn their customers' shopping behavior. Moreover, the information about the customers' activities inside the store, such as the hot and cold zones, is sent to a Li-Fi cloud for the store to track and analyze. The mobile application helps the customer to locate the products in the store [37], [38].

To use the application, shoppers only have to type the name of the product and the application will lead them to the product place on a displayed map of the store. Shoppers can also receive personalized notifications and coupons when they are near relevant products. The application tracks shoppers'

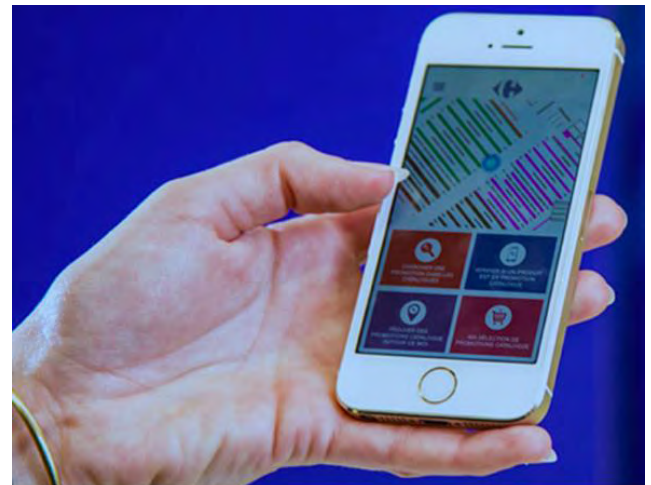


FIGURE 9. Main user interface of Carrefour mobile application [34].

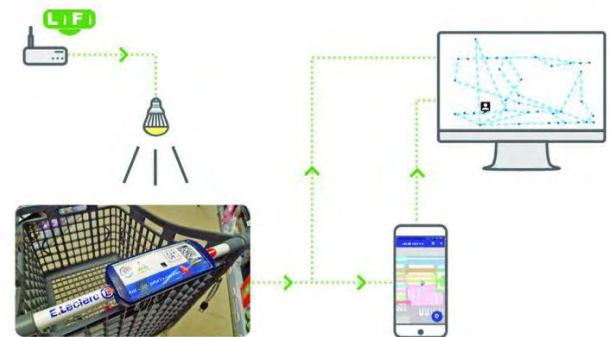


FIGURE 10. E.leclerc Li-Fi system [37].

behavior and highlights the area where they spend most of their time [39].

3) Grand Curtius Museum

Oledcomm company started a project with the Grand Curtius museum in Belgium, aiming to install a Li-Fi navigation system, creating a unique experience for the museum's visitors. The LEDs along the museum hall, as shown in Fig. 12, are equipped with Li-Fi technology to guide the visitors throughout their journey. The visitors no longer need to look for information about the artworks. With Li-Fi technology deployed, the visitors only need to stand under an LED close to a piece of art and the LiFi system will send the appropriate description to a special application installed in the museum's Li-Fi tablet. The application provides more capability by allowing the description to be a text, image, or even a video [40].

4) QUARTIER CAMILLE CLAUDEL

Camille Claudel is a mobile application developed to deliver geo-contextualize information. The application inhabitant in Palaiseau city, France, uses public lighting for information



FIGURE 11. E.leclerc mobile application UI [39].



FIGURE 14. Li-Fi luminaire solution [42].



FIGURE 12. Illustration of Li-Fi installed at Grand Curtius museum [40].

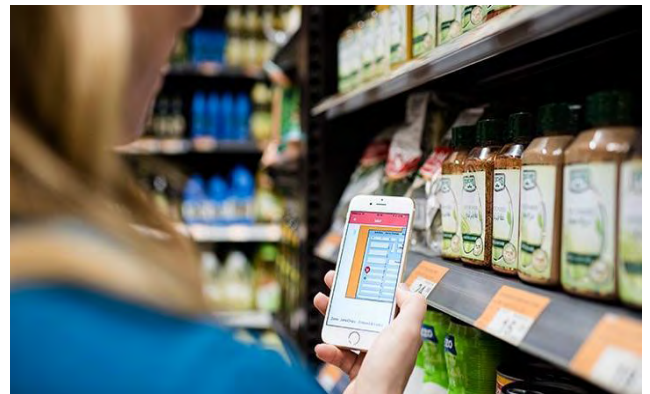


FIGURE 15. Aswaaq mobile application [44].

The application provides citizens with information about their area and their neighborhood. The application is available for both iOS and Android, and is developed by Oledcomm in cooperation with the Palaiseau city government [41].

5) EUROPE'S FIRST INDUSTRIALIZED LI-FI LUMINAIRE

In 2016, PureLiFi and Lucibel, a French company that designs and produces lighting solutions based on LED technology, cooperated in a project aiming to provide internet access through the light. The cooperation between the two companies resulted in the development of Europe's first industrialized Li-Fi luminaire, which supports bidirectional data transfers up to 42 Mbps. Using the Li-Fi luminaire with a LiFi-X dongle, which is a pureLiFi product, provides a bidirectional Li-Fi system. The Li-Fi luminaire acts as an access point connected to the backbone network using either POE or PLC. It has an infrared receiver used to detect the uplink from the dongle. The LiFi-X dongle connects to a mobile device or laptop through USB ports and contains a photoreceptor that receives a signal from the luminaire and an infrared transmitter that sends data back. This solution has been installed at multiple locations including Microsoft headquarters in Paris [42], [43].

6) ASWAAQ

Aswaaq is a mobile application that is developed to geolocate products in the Aswaaq retail store in Dubai. The store uses VLC-equipped LEDs to provide indoor positioning. The application enables the customers to build their shopping list in the app at home. When they arrive at the store, the application calculates the shortest route taken to

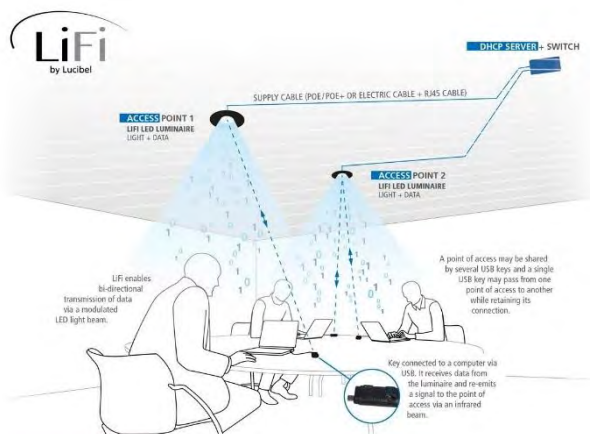


FIGURE 13. Li-Fi system developed by Lucibel [42].

exchange. The application is compatible with Oledcomm's geo-LiFi technology; therefore, the city installed 77 urban Li-Fi lightings equipped with geo-LiFi modules.

Palaiseau is the first city in France to experiment urban Li-Fi technology. The main goal of the project is to observe how people will use and engage with this technology.

TABLE 3. Application comparison.

Features	Applications			
	Quartier Camille Claudel	E.leclerc	Aswaaq	Carrefour Lille
Developing Company	Oledcomm	Oledcomm and E.Leclerc store	Philips, Aisle411, Value Labs, AlphaData and Aswaaq	Philips, IBA and Carrefour
Platform	Android and IOS	Android and IOS	Android and IOS	Android and IOS
Language	French	French	Arabic and English	French
Category	Smart City	Smart Building	Smart Building	Smart Building
User Account	X	✓	✓	✓
Manage content (add, edit, delete)	✓	✓	✓	✓
Manage LED information (add, edit, delete)	✓	✓	✓	✓
View statistics about the LEDs	X	✓	✓	✓
Features for admin				
View content	✓	✓	✓	✓
Save content	X	✓	✓	✓
Upload content	X	X	X	X
View content uploaded by users	X	X	X	X
Browse visual map of LED location	X	✓	✓	✓

complete the shopping list. Furthermore, it allows the admin to promote items for customers. These promotions pop-up during the shopping journey and can be personalized based on the customer's preferences and shopping history. The application can append products to the customer's shopping list based on several suggested recipes and their ingredients. The application is available for both iOS and Android, and has been developed by Philips in cooperation with Aisle411, ValueLabs, AlphaData, and Aswaaq stores [44].

Table 3 shows a comparison between the mobile applications developed by different companies to serve various customer needs. The criteria of comparison are chosen to reveal the gaps in the developed applications and signify what requires improvement.

According to the comparison between the applications presented in Table 3, most of the applications are used for navigation and customer guidance inside buildings. Furthermore, all applications lack an important feature as they do not give the user the ability to contribute in generating content and view other users' content. Providing the feature of content contribution by users allows system owners to collect huge contextualized data for analysis and improvement purposes.

Most of the applications are specific to particular functions in relation to the area in which they are deployed, such as retail store navigation or tourism. The content of the applications is generated and managed by system administrators. This shows a limitation in which end users cannot participate in the process of content generation, and therefore, creates a gap that needs to be filled to utilize huge contextual data collected from end users.

This can be a motivation to develop a Li-Fi-based IoT framework that uses light sources integrated with the Li-Fi technology to collect huge amounts of geo-contextualized data for further analysis and processing. This framework can be presented as a hierarchical architecture comprising multiple tiers: collection and aggregation of user data, processing, visualizing, and analyzing huge data, which can improve decision-making.

V. PROPOSED ARCHITECTURE

IoT, or in other words, "Network of Things," is the extension of internet connectivity beyond traditional devices [45]. The concept of IoT refers to the integration of objects into the virtual world, forming a network of connected intelligent things. Li-Fi is considered as an appealing IoT application

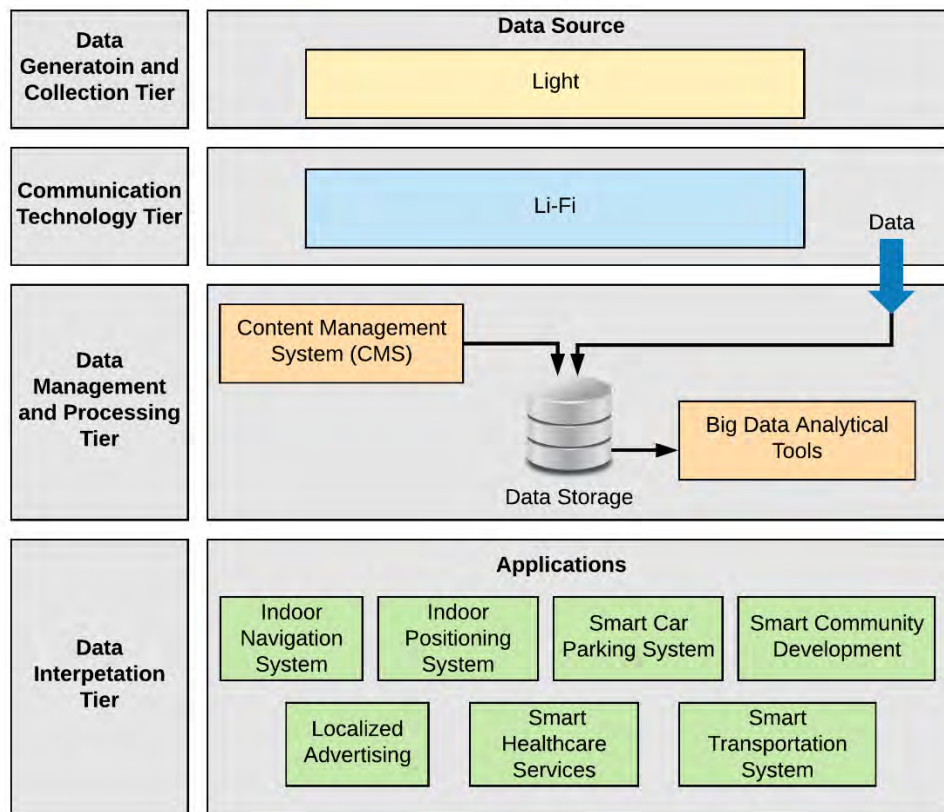


FIGURE 16. Proposed Li-Fi-based IoT hierarchical architecture.

since it is related to transforming light into an effective means of connectivity.

To generate data for analysis aiming at establishing a Li-Fi-based IoT system, an architecture is proposed to deploy Li-Fi in multiple indoor and outdoor environments. In the proposed architecture, the source of data dissemination and data collection is Li-Fi light. The collected data are analyzed and processed and a decision is performed based on the result.

The proposed architecture comprises four tiers: (1) data generation and collection, (2) communication technology, (3) data management and processing, and (4) data interpretation.

Fig. 16 illustrates the proposed layered architecture [46].

A. DATA GENERATION AND COLLECTION TIER

This tier concerns the collection of Li-Fi generated data. The source of the data is light empowered with Li-Fi technology. The essence of this tier is integrating Li-Fi in multiple indoor and outdoor locations and enables interaction with the technology with Li-Fi-compatible systems and applications to generate data and create a network of interconnected lights.

B. COMMUNICATION TECHNOLOGY TIER

The Li-Fi technology creates the means of communication in the proposed architecture. The tier is responsible for creating the communication means between the data source and the data storage hub. Li-Fi can be deployed in basic LED bulbs

to transmit signal information through the light. The Li-Fi lamps act as the end point of communication and can be connected to the LAN through basic internet infrastructure (i.e., Ethernet) or be assisted with other means of wireless communication technologies.

C. DATA MANAGEMENT AND PROCESSING TIER

The Li-Fi-generated data are collected, analyzed, and processed to extract geo-contextualized knowledge. Since Li-Fi can generate location-based content when deployed in particular areas, content management systems (CMSs) play an important role in the generation and management of location-related content. Data generated from users, things, and CMS are integrated in data storage for further processing. The utmost goal in this tier is to extract knowledge from the enormous data collected to help shaping smart services and applications. This tier acts as a mediator between the acquisition of data and application tier.

D. DATA INTERPRETATION TIER

The last tier is responsible for interpreting the processed data to make use of the result in many applications and decision-making.

Fig. 17 presents the various Li-Fi applications and their roles of data collection and generation. Li-Fi can be deployed in indoor environments such as hospitals, offices, museum,

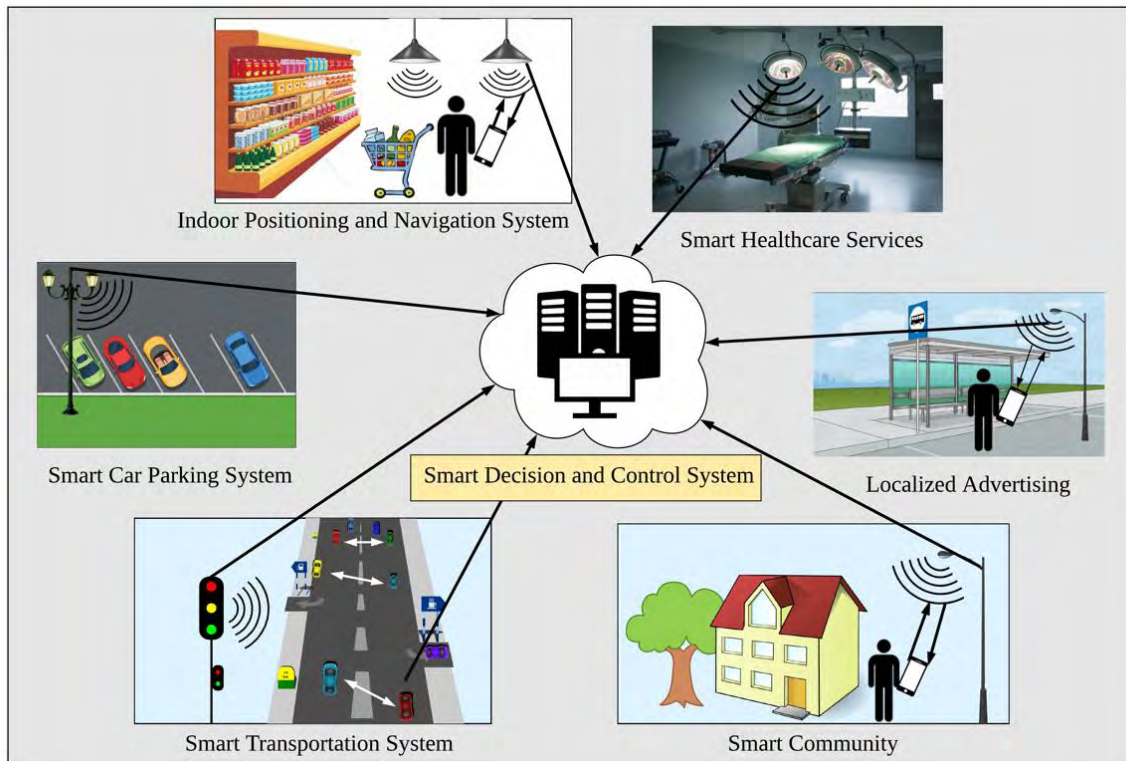


FIGURE 17. IoT system enabled by Li-Fi.

homes, and retail stores to create an indoor positioning and navigation system. Considerable data can be collected from the deployment of Li-Fi; when the data are analyzed, decision-makers can retain valuable information about these areas, such as hotspots, user's preferences, trends, and proximity advertising. Regarding smart city applications such as smart transportation systems, traffic control, parking systems, and smart street lighting, data generated using the Li-Fi technology can be beneficial to metropolis in order to enhance the urban services and create more livable, economically sound cities.

VI. CONCLUSION

Li-Fi is a visible light communication technology that utilizes LED bulbs for data transmission. Li-Fi has been receiving considerable attention of researchers, companies, and universities around the world. Currently, it is considered as a complementary technology to the current means of wireless communication. However, its importance is expected to grow drastically in the upcoming years. This highlights the need to draw attention into supporting the Li-Fi industrial growth and developing Li-Fi-compatible solutions that are suitable for use by people outside laboratory doors. Moreover, there is a lack of survey papers that thoroughly cover Li-Fi information, show how it evolved during the past years, and provide a review of the research and projects conducted on the technology. This paper tries to fill this gap by providing a

comprehensive survey on the Li-Fi technology, which includes background information required to understand it, in addition to a walkthrough of the research conducted on the technology to support its development. Various industrial efforts from companies globally have been highlighted to embrace Li-Fi-compatible solutions and products. In addition, this paper provides a Li-Fi-based IoT architecture, which is provided to create a tiered IoT system enabled by Li-Fi technology to aid shaping many Li-Fi applications in different sectors. This architecture is considered as a starting point into developing a scalable and generic GeoLiFi-based system for indoor and outdoor environments. The system aims to develop a CMS for Li-Fi-generated content and a mobile-based application for regular users. The first three tiers of the system are currently under development. When the system is deployed and fully functional, data will be collected and analyzed according to different scenarios. As the Li-Fi importance grows, the need for such systems will grow consequently.

For future works and to measure the performance and efficiency of the proposed system, it will be tested and evaluated in indoor and outdoor environments under various weather conditions and different scenarios (e.g. rainy). In addition, there are different potential research directions that can be considered. The security and privacy of Li-Fi technology is still an open issue. Also, it is recommended to study how to improve the coverage of LiFi technology by extending the

transmission range. Moreover, the attention should be given to the modulation techniques, which have been considered as the key of Li-Fi communication.

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